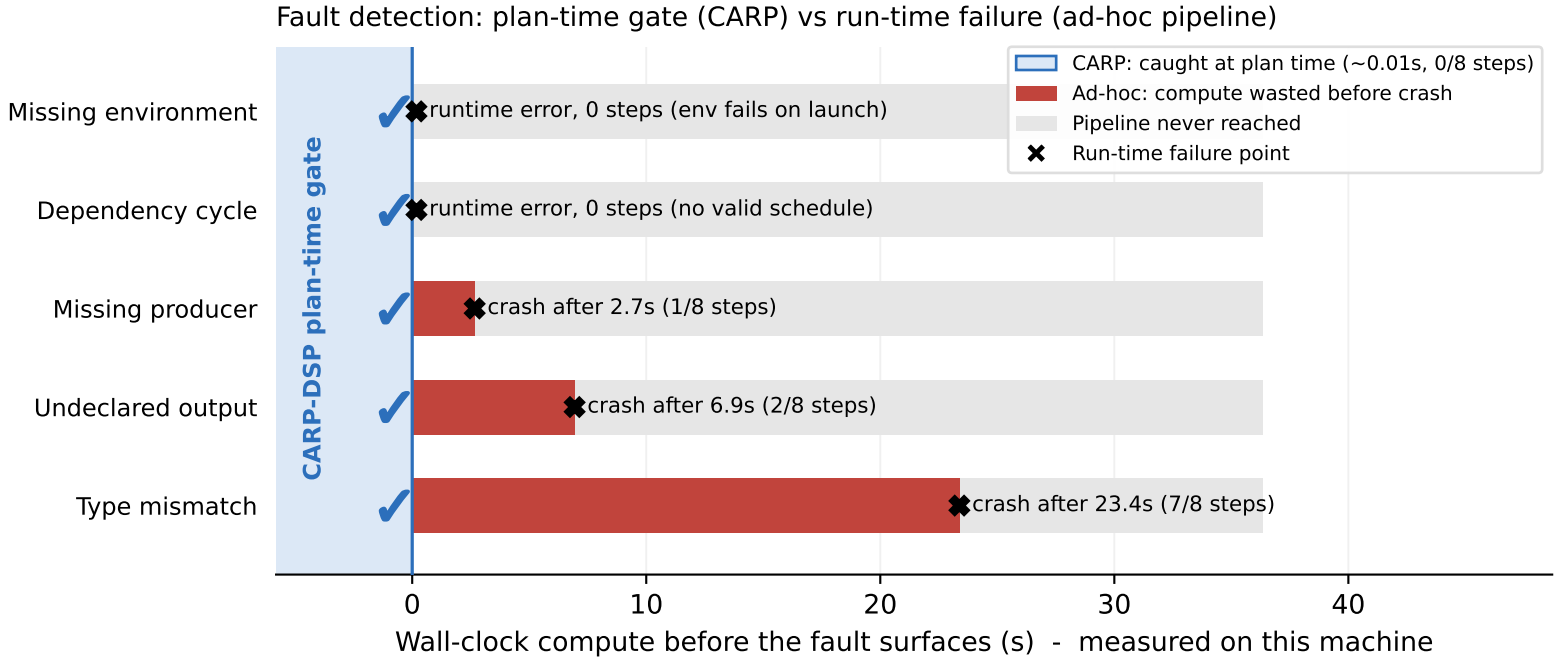
Fault detection: plan-time gate (CARP) vs run-time failure (ad-hoc pipeline)
CARP: caught at plan time (~0.01s, 0/8 steps)
Ad-hoc: compute wasted before crash
Pipeline never reached
Run-time failure point
CARP-DSP plan-time gate
Missing environment
Dependency cycle
Missing producer
Undeclared output
Type mismatch
runtime error, 0 steps (env fails on launch)
runtime error, 0 steps (no valid schedule)
crash after 2.7s (1/8 steps)
crash after 6.9s (2/8 steps)
crash after 23.4s (7/8 steps)
0
10
20
30
40
Wall-clock compute before the fault surfaces (s) - measured on this machine